

Supplementary Information

Allosteric Protein Chemical Shift Perturbations are Ubiquitous
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Table of Contents

Supplementary Text: The CSPdb

- SI Table S1. Data for all receptors in this study
- SI Table S2. Data for Hydrolase Receptors
- SI Fig S1. Confusion Matrix Histograms for Hydrolase Receptors
- SI Table S3. Data for Transferase Receptors
- SI Fig S2. Confusion Matrix Histograms for Transferase Receptors
- SI Table S4. Data for All Alpha Receptors
- SI Fig S3. Confusion Matrix Histograms for All Alpha Receptors
- SI Table S5. Data for All Beta Receptors
- SI Fig S4. Confusion Matrix Histograms for All Beta Receptors
- SI Table S6. Data for Alpha and Beta (a+b) Receptors
- SI Fig S5. Confusion Matrix Histograms for Alpha and Beta (a+b) Receptors
- SI Table S7. Data for BET-ET Domain Receptors
- SI Fig S6. Confusion Matrix Histograms for BET-ET Domain Receptors
- SI Table S8. Data for TFIIH Domain Receptors
- SI Fig S7. Confusion Matrix Histograms for TFIIH Domain Receptors
- SI Table S9. Data for Ubiquitin Domain Receptors
- SI Fig S8. Confusion Matrix Histograms for Ubiquitin Domain Receptors
- SI Table S10. Data for targets with dissimilar apo/holo experimental conditions
- SI Fig S9. Confusion Matrix Histograms for targets with dissimilar apo/holo experimental conditions
- SI Fig. S10. CA-inclusive CSP Confusion Matrix Histograms
- SI Fig. S11. CA-inclusive vs exclusive CSP F1 scores
- SI Fig. S12. CSP DB Confusion Matrix Histograms by closest interchain N-N distance
- SI Fig. S13. CSP DB Confusion Matrix Histograms by closest interchain atom-atom distance
- SI Fig. S14. F1 Scores for 1D CSPs
- SI Fig. S15. PDB Advanced Search Settings

SI Fig. S16. Impact of Experiment Condition Thresholding on Results
SI Fig. S17. Ideal N/H offsets
SI Fig. S18. CSP significance threshold Histogram
SI Fig. S19. F1 vs MCC scatterplot

References used in Supplementary Text

Supplementary Text

The CSPdb

The CSPdb includes receptors that regulate diverse biological processes, including cell cycle progression (e.g., MAD2: 1KLQ), transcriptional control (e.g., Nrd1: 2MOW, 6GC3), translational regulation, autophagy, apoptosis (e.g., calcineurin: 2JOG; BCL-XL family: 2M04, 2LP8), and cell adhesion (e.g., tensin: 2LOZ). Disease-relevant targets address mechanisms in leukemia and other cancers (e.g., p53: 2MEJ), Alzheimer's disease (e.g., amyloid precursor protein-binding proteins: 2ROZ), and infectious diseases from viruses and bacteria, such as anthrax (2RUI), chikungunya (5I22), SARS-CoV-2 (7PKU), HIV (2L6E), and Ebola (2KQ0).

The CSPdb features several recurring protein domains that exhibit notable conformational changes upon ligand binding. For instance, it includes 23 interactions involving calmodulin (CaM) or CaM-like proteins with peptides (e.g., PDB IDs: 1SY9, 2JZI; full list in **SI Table S1**), which often involve substantial rearrangements upon binding described in detail elsewhere(1). Other receptors known for binding-induced conformational shifts or folding include the G protein Cdc42 (1CF4), the metaphase chromosome spindle checkpoint protein MAD2 (1KLQ), the KIX domain (2LXS), and CREB-binding proteins (e.g., 1R8U, 1L8C).

Additional recurring domains include the extra-terminal (ET) domains of bromodomain proteins BRD3 and BRD4 (e.g., 2NCZ, 2ND0; full list in **SI Table S7, SI Figure 6**), which bind intrinsically-disordered regions of viral integrase or chromatin regulator proteins. These interactions trigger subtle allosteric changes distal from the binding site, including the formation of a stabilizing secondary structure that reorients two allosteric helices (2, 3). Similarly, the PDZ2 domain (e.g., 1D5G, 1VJ6, 2M0V) shows allosteric effects mediated by the β 2– β 3 loop during peptide recognition (4).

Nine entries involve the PH domain of the TFIIH transcription initiation factor subunit Tfb1 (e.g., 2LOX, 2M14; full list in **SI Table S8, SI Figure 7**). In these complexes, the ligand is often modeled as a helical polypeptide bound orthogonally to an extended β -sheet, resulting in propagation of allosteric signals via hydrogen-bond rearrangements. Other targets which involve hydrogen bond rearrangement in β -sheets include the Cdc42 (1CF4), Cyclophilin A (2MS4), and SH3 domain (e.g., 2OJ2, 2ROL; full list in **SI Table S1**). An additional eight entries use ubiquitin or ubiquitin-like (e.g. SUMO and NEDD8) protein receptors. Ubiquitin has been used as a model system for NMR studies (5) and for studying allostery (6) (e.g., 2KWV, 2MUR; full list in **SI Table S9, SI Figure 8**).

In Other Supplementary Information, you will find a All_Case_Studies.pdf, which provides a concise summary for each of the 139 targets in the CSPdb. Each case study contains the following panels: A) a $^{15}\text{N}/^1\text{H}$ HSQC scatterplot overlaying the apo and holo assigned spectra

with the optimal N/H offset applied; B) Calculated CSPs for receptor residues along the sequence, bars colored by confusion matrix classification; C) the mask of significant CSPs; D) the mask of binding-site residues; E) the confusion matrix classification projected onto the structure: allosteric CSPs are purple (FP), CSPs in the binding site are green (TP), non-CSPs in the binding site are orange (FN), and non-CSPs outside of the binding site are blue (TN). Residues without CSP measurements are shown in gray; ligands are colored cyan.